Elementary Analysis

Chapter 2, Section 11

Subsequences
Exercises 11.1–11.11, with every subpart

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Contents

11 Subsequences **2**

11.1 Exercise 11.1: A constant subsequence of a two-valued sequence 2
11.2 Exercise 11.2: Five questions for each of four elementary sequences 2
11.3 Exercise 11.3: Five questions for four bounded sequences 6
11.4 Exercise 11.4: Five questions for oscillation with and without growth 9
11.5 Exercise 11.5: Subsequential limits of an enumeration of a rational interval 12
11.6 Exercise 11.6: A subsequence of a subsequence 13
11.7 Exercise 11.7: A subsequence of an enumeration of all rational numbers 14
11.8 Exercise 11.8: Reflection exchanges the limit inferior and limit superior 15
11.9 Exercise 11.9: Closed intervals and an impossible set of subsequential limits 16
11.10 Exercise 11.10: The alternating diagonal traversal in Figure 11.2 17
11.11 Exercise 11.11: An increasing sequence approaching a supremum from the set . . . 19

11 Subsequences

A subsequence has the form $(s_{n_k})_{k\in\mathbb{N}}$, where its original indices satisfy $n_1 < n_2 < \cdots$. The selection map is $\sigma(k) = n_k$. The index k counts selected terms, whereas n_k is their position in the original sequence.

Subsequential limits in this section may be real numbers or $+\infty$ or $-\infty$. A proposed list must account for every possible subsequential limit. If S denotes the set of those limits, we use

$$\limsup s_n = \sup S, \qquad \liminf s_n = \inf S,$$

as established in Theorem 11.8. Increasing and decreasing remain weak notions, so a constant subsequence is monotone.

11.1 Exercise 11.1: A constant subsequence of a two-valued sequence

What we must show. For $a_n = 3+2(-1)^n$, list the first eight terms and give a constant subsequence together with its selection function.

Plan. Separate even and odd indices. The selection function must choose positive integer indices in strictly increasing order; specifying a repeated value without its indices is not enough.

(a) Compute the first eight terms.

At an odd index, $(-1)^n = -1$, so $a_n = 3 + 2(-1) = 3 - 2 = 1$. At an even index, $(-1)^n = 1$, so $a_n = 3 + 2(1) = 3 + 2 = 5$. Substituting the first eight indices gives

n	1	2	3	4	5	6	7	8
$(-1)^n$	-1	1	-1	1	-1	1	-1	1
a_n	1	5	1	5	1	5	1	5

(b) Specify and verify a selection function.

Define $\sigma : \mathbb{N} \to \mathbb{N}$ by $\sigma(k) = 2k$. For every $k \geq 1$, $\sigma(k) \geq 2 > 0$, and

$$\sigma(k+1) - \sigma(k) = 2(k+1) - 2k = 2 > 0.$$

Thus this is a valid strictly increasing sequence of indices. The selected values are

$$a_{\sigma(k)} = a_{2k} = 3 + 2(-1)^{2k} = 3 + 2 = 5 \qquad (k \in \mathbb{N}).$$

They form the constant subsequence $(5, 5, 5, \ldots)$.

Conclusion. The first eight terms are $1, 5, 1, 5, 1, 5, 1, 5$. One constant subsequence is $a_{2k} = 5$, selected by $\sigma(k) = 2k$. Selecting $\sigma(k) = 2k - 1$ would instead give the constant value 1.

11.2 Exercise 11.2: Five questions for each of four elementary sequences

What we must show. For each of

$$a_n = (-1)^n, \qquad b_n = \frac{1}{n}, \qquad c_n = n^2, \qquad d_n = \frac{6n+4}{7n-3},$$

answer all five questions: (a) a monotone subsequence with its indices; (b) the entire set of subsequential limits; (c) the limsup and liminf; (d) finite convergence or divergence to either infinity; and (e) boundedness. Subsequential limits here include the symbols $+\infty$ and $-\infty$ as in Definition 11.6.

Plan. We first record the tools used throughout Exercises 11.2–11.4. Each sequence then receives its own answers (a)–(e). In particular, constructing some subsequential limits will always be followed by a reason that no additional ones have been missed.

Tools shared by the three catalogues

Tool 1: Selection indices tend to infinity.

For any strictly increasing positive integer indices $n_1 < n_2 < \cdots$, one has $n_k \geq k$. At $k = 1$, $n_1 \geq 1$. If $n_k \geq k$, then the integer $n_{k+1} > n_k$ satisfies $n_{k+1} \geq n_k + 1 \geq k + 1$. Induction proves the claim. Consequently $k > N$ implies $n_k > N$.

The identity selection $n_k = k$ is allowed: a subsequence may retain all the terms of the original sequence. A constant subsequence is monotone under the book's non-strict definition, since its consecutive values satisfy both $u_{k+1} \geq u_k$ and $u_{k+1} \leq u_k$.

Tool 2: Every subsequence inherits an existing finite or infinite limit.

For a finite limit, this is Theorem 11.3. Its proof uses Tool 1: once every original index $n > N$ satisfies an error bound, every selected index n_k with $k > N$ satisfies it too. The same index argument proves the infinite cases directly. If $u_n \to +\infty$ and $M > 0$, choose N with $n > N \Longrightarrow u_n > M$. For $k > N$, $n_k \geq k > N$ gives $u_{n_k} > M$, so $u_{n_k} \to +\infty$. If $u_n \to -\infty$ and $M < 0$, choose N with $n > N \Longrightarrow u_n < M$; the identical implication $n_k \geq k > N$ gives $u_{n_k} < M$ and hence the negative infinite limit.

These alternatives are mutually exclusive. A finite limit L puts all sufficiently late terms in $(L-1, L+1)$, which prevents either infinite limit. The two signs of infinity are incompatible because their threshold tests with 1 and -1 would require a late term to be both greater than 1 and less than -1. Finite limits are unique by Theorem 7.3. Thus, if the original sequence has any existing limit L, its set of subsequential limits is exactly $\{L\}$: every subsequence has that limit, and the identity selection itself supplies a subsequence.

Tool 3: A sequence with only finitely many values.

Suppose every term belongs to a finite set $F = \{f_1, \ldots, f_m\}$ and every member of F occurs infinitely often. For each particular sequence below, we will display constant selections proving that every member of F is a subsequential limit. There can be no other finite subsequential limit. Indeed, if $L \notin F$, then the finite set of positive distances $\{|L - f_i| : 1 \leq i \leq m\}$ has a positive minimum δ. Every term, including every selected term, satisfies $|u_n - L| \geq \delta$, so convergence to L fails the error test with $\varepsilon = \delta/2$.

There are no infinite subsequential limits either. With $C = \max\{|f_1|, \ldots, |f_m|\}$, every term lies between $-C$ and C. Neither the positive threshold $C + 1$ nor the negative threshold $-C - 1$ can ever be crossed in the required direction. Therefore the full subsequential-limit set is precisely F.

Tool 4: Extract limsup, liminf, and full-sequence behavior from the set.

If S is the full set of subsequential limits, Theorem 11.8(ii) gives

$$\limsup_{n\to\infty} u_n = \sup S, \qquad \liminf_{n\to\infty} u_n = \inf S.$$

In the explicit sets below these are their greatest and least elements in the extended order. Theorem 11.8(iii) says that the original sequence has a limit if and only if S has exactly one element. A singleton real element means finite convergence; a singleton $\{+\infty\}$ or $\{-\infty\}$ means divergence of the corresponding sign. If S contains two or more elements, there is no finite limit and neither infinite limit exists.

The sequence $a_n = (-1)^n$

(a) A monotone subsequence. Choose $n_k = 2k$. The indices start at 2 and increase by 2, so they are positive and strictly increasing. Then $a_{n_k} = (-1)^{2k} = 1$ at every k. This constant subsequence is monotone.

(b) All subsequential limits. At the indices $2k-1$ the values are -1, and at the indices $2k$ they are 1. Both selections have positive indices and successive differences 2. Thus constant subsequences give both -1 and 1 as limits. The original sequence takes only these two values, so Tool 3 excludes every other finite or infinite subsequential limit:

$$S_a = \{-1, 1\}.$$

(c) Limsup and liminf. The greatest element of S_a is 1 and the least is -1. Tool 4 therefore gives $\limsup a_n = 1$ and $\liminf a_n = -1$.

(d) Full-sequence behavior. Since S_a has two distinct elements, Tool 4 excludes all three possibilities: (a_n) does not converge to a real number, does not diverge to $+\infty$, and does not diverge to $-\infty$.

(e) Boundedness. Every term is either -1 or 1, so $-1 \le a_n \le 1$ for every n. Thus (a_n) is bounded.

The sequence $b_n = 1/n$

(a) A monotone subsequence. Take the identity selection $n_k = k$, whose indices start at 1 and increase by 1. It gives $b_{n_k} = 1/k$. Its consecutive differences are

$$\frac{1}{k+1} - \frac{1}{k} = \frac{k-(k+1)}{k(k+1)} = -\frac{1}{k(k+1)} < 0.$$

The denominator is positive for $k \ge 1$, so the selected sequence is strictly decreasing.

(b) All subsequential limits. The original sequence converges to 0: given $\varepsilon > 0$, $n > 1/\varepsilon$ implies $|1/n - 0| = 1/n < \varepsilon$. Tool 2 therefore proves that every subsequence has limit 0, and excludes any additional finite or infinite limit. Thus $S_b = \{0\}$.

(c) Limsup and liminf. The supremum and infimum of $\{0\}$ are both 0. Hence $\limsup b_n = \liminf b_n = 0$ by Tool 4.

(d) Full-sequence behavior. The preceding error estimate proves finite convergence to 0. It does not diverge to either sign of infinity, by the mutual exclusivity recorded in Tool 2.

(e) Boundedness. For every positive integer n, $0 < 1/n \le 1$. Thus $0 < b_n \le 1$, providing a lower and an upper bound.

The sequence $c_n = n^2$

(a) A monotone subsequence. Take $n_k = k$. Then $c_{n_k} = k^2$, and

$$(k+1)^2 - k^2 = k^2 + 2k + 1 - k^2 = 2k + 1 > 0.$$

Thus the identity subsequence is strictly increasing.

(b) All subsequential limits. The original sequence has limit $+\infty$: for $M > 0$, $n > \sqrt{M}$ implies $n^2 > M$. Tool 2 proves that every subsequence has this same infinite limit. Consequently the exact set is $S_c = \{+\infty\}$; in particular, no finite limit or negative-infinity limit has been omitted.

(c) Limsup and liminf. Both the greatest and least elements of this singleton are $+\infty$. Thus $\limsup c_n = \liminf c_n = +\infty$.

(d) Full-sequence behavior. The sequence diverges to $+\infty$. It has no finite real limit and does not diverge to $-\infty$.

(e) Boundedness. It is bounded below by 1, since $n \geq 1$. It is not bounded above: for any proposed upper bound B, choose $n > \max\{1, B\}$. Then $n^2 \geq n > B$. Thus (c_n) is not a bounded sequence.

The sequence $d_n = (6n + 4)/(7n - 3)$

For every $n \geq 1$, $7n - 3 \geq 4 > 0$, so each term is defined with a positive denominator. An exact subtraction identifies both its limit and its side of approach:

$$\begin{aligned} d_n - \frac{6}{7} &= \frac{7(6n+4) - 6(7n-3)}{7(7n-3)} \\ &= \frac{42n + 28 - 42n + 18}{7(7n-3)} \\ &= \frac{46}{7(7n-3)} > 0. \end{aligned}$$

(a) A monotone subsequence. Use $n_k = k$. Subtract consecutive values using the error formula just established:

$$\begin{aligned} d_{k+1} - d_k &= \frac{46}{7}\left(\frac{1}{7k+4} - \frac{1}{7k-3}\right) \\ &= \frac{46}{7}\,\frac{(7k-3) - (7k+4)}{(7k+4)(7k-3)} \\ &= \frac{46}{7}\,\frac{-7}{(7k+4)(7k-3)} \\ &= -\frac{46}{(7k+4)(7k-3)} < 0. \end{aligned}$$

Both denominator factors are positive for $k \geq 1$. Thus the full sequence, and hence the identity subsequence, is strictly decreasing.

(b) All subsequential limits. For $n \geq 1$, $7n - 3 \geq 4n$ because $(7n - 3) - 4n = 3(n - 1) \geq 0$. Therefore

$$\left| d_n - \frac{6}{7} \right| = \frac{46}{7(7n-3)} \leq \frac{46}{28n} = \frac{23}{14n}.$$

For $\varepsilon > 0$, any threshold $N > 23/(14\varepsilon)$ makes this error less than ε for $n > N$. Thus $d_n \to 6/7$. Tool 2 supplies exhaustiveness, so $S_d = \{6/7\}$.

(c) Limsup and liminf. Tool 4 applied to the singleton gives $\limsup d_n = \liminf d_n = 6/7$.

(d) Full-sequence behavior. The sequence converges to the finite value $6/7$. Neither sign of infinite divergence is possible.

(e) Boundedness. The positive error formula shows $d_n > 6/7$ for every n. Strict decrease shows no term is larger than the first:

$$d_1 = \frac{6+4}{7-3} = \frac{10}{4} = \frac{5}{2}.$$

Thus $6/7 < d_n \leq 5/2$ for every n, so the sequence is bounded.

Conclusion. In Exercise 11.2, (b_n) and (d_n) converge, to 0 and $6/7$ respectively; (c_n) diverges to $+\infty$; and (a_n) has no extended limit. None diverges to $-\infty$. Exactly $(a_n), (b_n), (d_n)$ are bounded. Each of the four sequences has been given a monotone selection and its complete subsequential-limit set above.

11.3 Exercise 11.3: Five questions for four bounded sequences

What we must show. Answer (a)–(e) of Exercise 11.2 separately for

$$s_n = \cos(n\pi/3), \qquad t_n = \frac{3}{4n+1}, \qquad u_n = \left(-\frac{1}{2}\right)^n, \qquad v_n = (-1)^n + \frac{1}{n}.$$

We continue to use the four shared tools, so all sets of subsequential limits are understood in the extended sense.

The sequence $s_n = \cos(n\pi/3)$

The cosine values are determined by the residue of n modulo 6. Write $n = 6q + r$, with $q \geq 0$ an integer and $r \in \{1, 2, 3, 4, 5, 6\}$. Then $n\pi/3 = 2q\pi + r\pi/3$, so the period 2π gives

r	1	2	3	4	5	6
$\cos(r\pi/3)$	$1/2$	$-1/2$	-1	$-1/2$	$1/2$	1

(a) A monotone subsequence. Select $n_k = 6k$. These positive indices start at 6 and increase by 6. The selected terms satisfy $s_{6k} = \cos(2k\pi) = 1$, so this is a constant monotone subsequence.

(b) All subsequential limits. The table gives precisely four different values. The following explicit selections show that each occurs as a constant subsequence:

indices n_k	first index	s_{n_k}
$6k-5$	1	$1/2$
$6k-4$	2	$-1/2$
$6k-3$	3	-1
$6k$	6	1

Each first index is positive and each sequence of indices increases by 6. The period table proves there are no other values; Tool 3 excludes any other finite or infinite subsequential limit. Hence

$$S_s = \{-1, -1/2, 1/2, 1\}.$$

(c) Limsup and liminf. The greatest element is 1 and the least is -1, so $\limsup s_n = 1$ and $\liminf s_n = -1$.

(d) Full-sequence behavior. The set S_s has four distinct elements, rather than one. Tool 4 shows that the original sequence does not converge to a finite value and does not diverge to either infinity.

(e) Boundedness. The table puts every value between -1 and 1, so $-1 \le s_n \le 1$ for every n and the sequence is bounded.

The sequence $t_n = 3/(4n+1)$

(a) A monotone subsequence. The denominators are positive because $4n+1 \ge 5$. Choose $n_k = k$. The difference between consecutive selected terms is

$$\begin{aligned} t_{k+1} - t_k &= \frac{3}{4k+5} - \frac{3}{4k+1} \\ &= \frac{3[(4k+1)-(4k+5)]}{(4k+5)(4k+1)} \\ &= -\frac{12}{(4k+5)(4k+1)} < 0. \end{aligned}$$

Thus the identity subsequence is strictly decreasing.

(b) All subsequential limits. For every $n \ge 1$,

$$0 < t_n = \frac{3}{4n+1} \le \frac{3}{4n}.$$

For $\varepsilon > 0$, $n > 3/(4\varepsilon)$ makes the last expression less than ε. Hence $t_n \to 0$. Tool 2 proves that all subsequences have the same limit, so $S_t = \{0\}$, with no additional extended limits.

(c) Limsup and liminf. Both extremal elements of this singleton are 0. Thus $\limsup t_n = \liminf t_n = 0$.

(d) Full-sequence behavior. The sequence converges to 0 and diverges to neither $+\infty$ nor $-\infty$.

(e) Boundedness. The inequality $4n+1 \ge 5$ gives $0 < t_n \le 3/5$. Therefore the sequence is bounded.

The sequence $u_n = (-1/2)^n$

(a) A monotone subsequence. Choose the even indices $n_k = 2k$, positive and increasing by 2. Then

$$u_{2k} = \left(-\frac{1}{2}\right)^{2k} = \left(\frac{1}{4}\right)^k > 0.$$

The next selected term is $u_{2(k+1)} = (1/4)u_{2k}$. Since $0 < 1/4 < 1$ and $u_{2k} > 0$, it is strictly smaller than u_{2k}. Thus this subsequence is strictly decreasing.

(b) All subsequential limits. The base has magnitude $1/2 < 1$. Theorem 9.7(b), or Exercise 9.13, gives $(-1/2)^n \to 0$. Equivalently its distance from 0 is $(1/2)^n \to 0$, independently of the alternating sign. Tool 2 then gives the exhaustive set $S_u = \{0\}$.

(c) Limsup and liminf. The singleton has both supremum and infimum 0, so $\limsup u_n = \liminf u_n = 0$.

(d) Full-sequence behavior. It converges to the finite value 0, and diverges to neither sign of infinity.

(e) Boundedness. For $n \geq 1$,

$$|u_n| = \left(\frac{1}{2}\right)^n = \frac{1}{2}\left(\frac{1}{2}\right)^{n-1} \leq \frac{1}{2}.$$

The final product is at most $1/2$ because its remaining $n-1$ factors are at most 1; at $n=1$ that remaining product is empty and equals 1. Hence $-1/2 \leq u_n \leq 1/2$, a bound valid at every index.

The sequence $v_n = (-1)^n + 1/n$

(a) A monotone subsequence. Choose $n_k = 2k$, again positive and increasing by 2. The selected terms are $v_{2k} = 1 + 1/(2k)$, and

$$\begin{aligned} v_{2(k+1)} - v_{2k} &= \frac{1}{2k+2} - \frac{1}{2k} \\ &= \frac{2k - (2k+2)}{(2k+2)(2k)} \\ &= -\frac{1}{2k(k+1)} < 0. \end{aligned}$$

This is a strictly decreasing subsequence.

(b) All subsequential limits. The even and odd selections respectively give

$$v_{2k} = 1 + \frac{1}{2k} \longrightarrow 1, \qquad v_{2k-1} = -1 + \frac{1}{2k-1} \longrightarrow -1.$$

Both index formulas start at positive integers and increase by 2. The reciprocal errors tend to 0: for example, $1/(2k) \leq 1/k$ and $1/(2k-1) \leq 1/k$ for $k \geq 1$. Thus -1 and 1 are both subsequential limits.

To exclude every other finite candidate, let $L \notin \{-1, 1\}$ and set

$$\delta = \min\{|L-1|, |L+1|\} > 0.$$

Consider any subsequence with indices n_k. Since $n_k \geq k$, we have $1/n_k \leq 1/k$. For $k > 2/\delta$, this is less than $\delta/2$. The reverse triangle inequality then yields

$$\begin{aligned} |v_{n_k} - L| &= |((-1)^{n_k} - L) + 1/n_k| \\ &\geq |(-1)^{n_k} - L| - \frac{1}{n_k} \\ &\geq \delta - \frac{1}{n_k} > \frac{\delta}{2}. \end{aligned}$$

This defeats the convergence test with error tolerance $\delta/2$ for every such subsequence. Finally,

$$|v_n| \leq |(-1)^n| + \frac{1}{n} = 1 + \frac{1}{n} \leq 2,$$

so no subsequence can tend to either infinity. The complete set is therefore $S_v = \{-1, 1\}$.

(c) Limsup and liminf. Tool 4 gives $\limsup v_n = 1$ and $\liminf v_n = -1$. The values of the sequence approach these numbers along the even and odd selections, respectively.

(d) Full-sequence behavior. There are two distinct subsequential limits. Thus the full sequence has no finite limit and diverges to neither infinity.

(e) Boundedness. The absolute-value estimate $|v_n| \le 2$ already proves boundedness. A sharper bound follows directly from parity: at even indices $n = 2k$, $1 < v_{2k} = 1 + 1/(2k) \le 3/2$; at odd indices $n = 2k-1$, $-1 < v_{2k-1} = -1 + 1/(2k-1) \le 0$. Combining the cases gives $-1 < v_n \le 3/2$ for every n.

Conclusion. In Exercise 11.3 all four sequences are bounded. The sequences (t_n) and (u_n) converge to 0; (s_n) and (v_n) have no extended limit. None diverges to either infinity. Their exact subsequential-limit sets are, in order, $\{-1, -1/2, 1/2, 1\}$, $\{0\}$, $\{0\}$, and $\{-1, 1\}$, with limsup and liminf the largest and smallest elements of these sets.

11.4 Exercise 11.4: Five questions for oscillation with and without growth

What we must show. Answer (a)–(e) of Exercise 11.2 separately for

$$w_n = (-2)^n, \qquad x_n = 5^{(-1)^n}, \qquad y_n = 1 + (-1)^n, \qquad z_n = n\cos(n\pi/4).$$

In the second expression, $(-1)^n$ is the *exponent* of 5; the expression is not multiplication by $(-1)^n$.

Plan. Parity resolves the first three sequences. For the last, a complete eight-residue table identifies the zero terms and the growing positive and negative terms. A separate bounded-tail argument then excludes any unlisted finite subsequential limit.

The sequence $w_n = (-2)^n$

(a) A monotone subsequence. Take $n_k = 2k$, whose positive indices increase by 2. Then $w_{2k} = 4^k > 0$. The next selected value is $4^{k+1} = 4 \cdot 4^k > 4^k$, so this subsequence is strictly increasing.

(b) All subsequential limits. The even selection gives $w_{2k} = 4^k \to +\infty$ by Exercise 9.13, since the base 4 is greater than 1. The odd selection $n_k = 2k - 1$ gives

$$w_{2k-1} = -2^{2k-1} = -\frac{1}{2}\,4^k \longrightarrow -\infty$$

by the negative scaling result of Exercise 9.10. Both selections are valid positive increasing indices.

There is no finite subsequential limit. Indeed, $|w_n| = 2^n \to +\infty$ by Exercise 9.13. Tool 2 applied to this positive magnitude sequence gives $|w_{n_k}| \to +\infty$ for every selection. If $w_{n_k} \to L \in \mathbb{R}$, the error tolerance 1 would give, eventually,

$$|w_{n_k}| \le |w_{n_k} - L| + |L| < 1 + |L|\,,$$

contradicting that magnitude limit. The only remaining extended candidates are the two already constructed, so

$$S_w = \{-\infty, +\infty\}.$$

(c) Limsup and liminf. The extremal elements of S_w give $\limsup w_n = +\infty$ and $\liminf w_n = -\infty$.

(d) Full-sequence behavior. Since S_w has two elements, the full sequence has no extended limit. In particular it does not converge to a real number and does not diverge to either single sign of infinity. Its large positive and negative terms do not constitute a limit of one sign.

(e) Boundedness. The even terms exceed every real upper bound and the odd terms go below every real lower bound, by their infinite limits just proved. Hence (w_n) is bounded neither above nor below, and is not a bounded sequence.

The sequence $x_n = 5^{(-1)^n}$

At even indices the exponent equals 1, and at odd indices it equals -1. Therefore

$$x_{2k} = 5^1 = 5, \qquad x_{2k-1} = 5^{-1} = \frac{1}{5}.$$

(a) A monotone subsequence. Choose $n_k = 2k$, positive and increasing by 2. Then $x_{n_k} = 5$ at every index, a constant monotone subsequence.

(b) All subsequential limits. The even and odd index selections are constant with values 5 and $1/5$ respectively, so both are subsequential limits. They are the only values the original sequence ever takes; Tool 3 therefore excludes all other finite and infinite subsequential limits. Thus $S_x = \{1/5, 5\}$.

(c) Limsup and liminf. Since $1/5 < 5$, Tool 4 gives $\limsup x_n = 5$ and $\liminf x_n = 1/5$.

(d) Full-sequence behavior. The two distinct subsequential limits show that (x_n) does not converge to a finite number and diverges to neither sign of infinity.

(e) Boundedness. Every term is either $1/5$ or 5, so $1/5 \le x_n \le 5$. Both bounding values are attained, and the sequence is bounded.

The sequence $y_n = 1 + (-1)^n$

(a) A monotone subsequence. Select $n_k = 2k$, positive and increasing by 2. Then $y_{2k} = 1 + 1 = 2$ at every selected index. This constant subsequence is monotone.

(b) All subsequential limits. At odd indices, $y_{2k-1} = 1 - 1 = 0$, while at even indices $y_{2k} = 2$. The two valid selections give constant subsequences with limits 0 and 2. Because these are the only values taken by any term, Tool 3 proves the exhaustive set $S_y = \{0, 2\}$, with no infinite subsequential limit.

(c) Limsup and liminf. The largest and smallest elements give $\limsup y_n = 2$ and $\liminf y_n = 0$.

(d) Full-sequence behavior. The set of subsequential limits has two distinct elements. Thus there is no finite limit and no divergence to either $+\infty$ or $-\infty$.

(e) Boundedness. The values satisfy $0 \le y_n \le 2$ at every index, so the sequence is bounded.

The sequence $z_n = n\cos(n\pi/4)$

Write $n = 8q + r$, with $q \ge 0$ an integer and $r \in \{1, 2, 3, 4, 5, 6, 7, 8\}$. Then $n\pi/4 = 2q\pi + r\pi/4$. The cosine period and its unit-circle values give the complete table

r	$\cos(r\pi/4)$	z_n when $n = 8q + r$
1	$1/\sqrt{2}$	$n/\sqrt{2}$
2	0	0
3	$-1/\sqrt{2}$	$-n/\sqrt{2}$
4	-1	$-n$
5	$-1/\sqrt{2}$	$-n/\sqrt{2}$
6	0	0
7	$1/\sqrt{2}$	$n/\sqrt{2}$
8	1	n

Thus every nonzero term has magnitude at least $n/\sqrt{2}$; the only other possibility is the exact value 0.

(a) A monotone subsequence. Take $n_k = 8k$, with first index 8 and successive difference $8 > 0$. The selected values are

$$z_{8k} = 8k\cos(2k\pi) = 8k.$$

Their consecutive differences equal $8 > 0$, so this is a strictly increasing subsequence.

(b) Construct the three candidate limits. The same even-period selection gives $z_{8k} = 8k \to +\infty$: given $M > 0$, every $k > M/8$ makes $8k > M$. For negative infinity, choose $n_k = 8k - 4$, starting at 4 and increasing by 8. Then

$$z_{8k-4} = (8k-4)\cos((2k-1)\pi) = -(8k-4) = -8k + 4.$$

For $M < 0$, every $k > (4-M)/8$ satisfies $-8k + 4 < M$, so this subsequence tends to $-\infty$.

For the finite value 0, choose $n_k = 4k - 2$, starting at 2 and increasing by 4. The angle is $(4k-2)\pi/4 = k\pi - \pi/2$, whose cosine is 0. Thus $z_{4k-2} = 0$ for every k, giving a constant subsequence with limit 0.

(b) Exclude every other finite limit. Suppose a subsequence z_{n_k} converges to a finite $L \in \mathbb{R}$. Convergence with error tolerance 1 gives a threshold K_1 such that for $k > K_1$,

$$|z_{n_k}| \le |z_{n_k} - L| + |L| < 1 + |L|\,.$$

Choose $K_2 > \sqrt{2}(1 + |L|)$. For $k > K_2$, Tool 1 gives $n_k \ge k > \sqrt{2}(1 + |L|)$, so

$$\frac{n_k}{\sqrt{2}} > 1 + |L|\,.$$

At an index with $k > \max\{K_1, K_2\}$, a nonzero selected term would have, by the period table, $|z_{n_k}| \ge n_k/\sqrt{2} > 1 + |L|$, contradicting the bounded-tail estimate. Thus every sufficiently late selected term must be exactly 0.

Such a subsequence has limit 0 directly from the definition. It cannot have a different finite limit $L \neq 0$, because its late errors from L would all equal $|L|$, failing the tolerance $\varepsilon = |L|/2$. Consequently 0 is the only possible finite subsequential limit. The only infinite possibilities, $+\infty$ and $-\infty$, have already been constructed. The complete set is

$$S_z = \{-\infty, 0, +\infty\}.$$

(c) Limsup and liminf. The greatest and least elements of this set are $+\infty$ and $-\infty$. Hence $\limsup z_n = +\infty$ and $\liminf z_n = -\infty$.

(d) Full-sequence behavior. There are three distinct subsequential limits. By Tool 4 the original sequence does not converge to a finite real number and does not diverge to either infinity. In particular, the growing magnitudes in some residue classes do not remove the infinitely many exact zero terms.

(e) Boundedness. The values $z_{8k} = 8k$ are unbounded above, and the values $z_{8k-4} = -8k + 4$ are unbounded below. Thus (z_n) is bounded neither above nor below, and is not a bounded sequence.

Conclusion. In Exercise 11.4 none of the four full sequences has a finite or infinite limit. The sequences (x_n) and (y_n) are bounded, while (w_n) and (z_n) are unbounded above and below. Their complete subsequential-limit sets are, respectively,

$$\{-\infty, +\infty\}, \qquad \{1/5, 5\}, \qquad \{0, 2\}, \qquad \{-\infty, 0, +\infty\}.$$

The limsup and liminf for each sequence are the greatest and least elements of its displayed set, as proved in the individual answers to (c).

11.5 Exercise 11.5: Subsequential limits of an enumeration of a rational interval

What we must show. Let (q_n) enumerate all rational numbers in $(0, 1]$. Find (a) its set of subsequential limits and (b) its limit superior and limit inferior.

Plan. Every term lies between 0 and 1, so a subsequential limit must lie in $[0, 1]$. Conversely, every neighborhood of a point of $[0, 1]$ contains infinitely many distinct rational values in $(0, 1]$. Removing a finite prefix of the enumeration cannot remove all those values. This will allow a construction with strictly increasing indices.

1. Explain the role of enumeration and distinct values

The enumeration hypothesis says that each rational number in $(0, 1]$ occurs somewhere in the list. It does not require a particular rational number to occur infinitely often. For example, the value 1 may occur only once. The proof below uses infinitely many *different* rational values near a proposed limit, so it works both with and without repeated terms.

Every nonempty open interval (c, d) contains infinitely many rational numbers. To prove this from density, suppose its rational numbers formed a finite set. Density ensures the set is nonempty, so it would have a largest member $r < d$. Density applied to (r, d) would produce a rational number larger than r still belonging to (c, d), contradicting the choice of r. Thus there are infinitely many distinct rational numbers in the interval.

2. Part (a): Find infinitely many eligible values near every target

Fix $t \in [0, 1]$ and $\varepsilon > 0$. Consider

$$c = \max\{0, t - \varepsilon\}, \qquad d = \min\{1, t + \varepsilon\}.$$

We have $c < d$. At $t = 0$, this follows from $c = 0 < \min\{1, \varepsilon\} = d$. At $t = 1$, it follows from $c = \max\{0, 1 - \varepsilon\} < 1 = d$. For $0 < t < 1$, both $0 < t$ and $t - \varepsilon < t$ give $c < t$, while $t < 1$ and $t < t + \varepsilon$ give $t < d$.

Every rational number $r \in (c, d)$ satisfies

$$0 < r < 1, \qquad t - \varepsilon < r < t + \varepsilon.$$

It therefore belongs to the enumerated set and satisfies $|r - t| < \varepsilon$. By Step 1, there are infinitely many distinct such rational values. A finite prefix $q_1, \ldots, q_N$ contains only finitely many values, so it cannot contain them all. Hence for every finite index bound N there is an index $n > N$ such that

$$|q_n - t| < \varepsilon.$$

This reasoning includes the endpoints $t = 0$ and $t = 1$.

3. Construct a subsequence converging to the chosen target

Put $n_0 = 0$. At stage $k \geq 1$, apply Step 2 with $\varepsilon = 1/k$ and with the finite index bound n_{k-1}. Choose an index $n_k > n_{k-1}$ satisfying

$$|q_{n_k} - t| < \frac{1}{k}.$$

For definiteness, one may take the least eligible index, since a nonempty set of positive integers has a least element. The inequalities $n_1 < n_2 < \cdots$ ensure that (q_{n_k}) is a subsequence.

Let $\eta > 0$ and choose an integer $K > 1/\eta$ by the Archimedean property. If $k > K$, then

$$|q_{n_k} - t| < \frac{1}{k} < \eta.$$

Thus $q_{n_k} \to t$. Since $t \in [0,1]$ was arbitrary, every point of that interval is a subsequential limit. This is also the infinite-neighborhood criterion of Theorem 11.2(i).

4. Exclude every other finite or infinite limit

Every subsequence still satisfies $0 < q_{n_k} \leq 1$. If it had a finite limit $t < 0$, use the positive tolerance $\eta = -t/2$. Eventually its terms would be less than $t + \eta = t/2 < 0$, a contradiction. If it had a finite limit $t > 1$, use $\eta = (t-1)/2$. Eventually its terms would be greater than $t - \eta = (t+1)/2 > 1$, again a contradiction. Hence every finite subsequential limit lies in $[0,1]$.

No subsequence can tend to $+\infty$, because its terms never exceed 1 and so cannot eventually exceed the positive threshold 2. No subsequence can tend to $-\infty$, because its terms are positive and so cannot eventually be less than -1. Combining this exclusion with Step 3 proves that the exact set of subsequential limits is $[0,1]$.

5. Part (b): Take the extreme subsequential limits

By Theorem 11.8(ii), the limit superior is the supremum of the set of subsequential limits, and the limit inferior is its infimum. The interval $[0,1]$ has maximum 1 and minimum 0, so

$$\limsup_{n\to\infty} q_n = 1, \qquad \liminf_{n\to\infty} q_n = 0.$$

Conclusion. (a) The set of subsequential limits is $[0,1]$. (b) The limit superior is 1 and the limit inferior is 0. No repeated occurrence of an individual rational value is assumed.

11.6 Exercise 11.6: A subsequence of a subsequence

What we must show. Prove that a subsequence of a subsequence of a given sequence is itself a subsequence of the original sequence.

Plan. Use Definition 11.1 in terms of strictly increasing selection functions. Compose the two selection functions and verify that the composition remains strictly increasing.

1. Write the two selections precisely

Let $s : \mathbb{N} \to \mathbb{R}$ represent the original sequence. Suppose t is a subsequence of s. There is a strictly increasing function $\sigma : \mathbb{N} \to \mathbb{N}$ with

$$t_k = s_{\sigma(k)} \qquad (k \in \mathbb{N}).$$

Now suppose u is a subsequence of t. There is a strictly increasing function $\tau : \mathbb{N} \to \mathbb{N}$ with

$$u_j = t_{\tau(j)} \qquad (j \in \mathbb{N}).$$

Here strictly increasing refers to the *indices* selected by the functions; the numerical sequence values themselves need not increase or be distinct.

2. Form and check the composite selection

Define $\rho = \sigma \circ \tau$; explicitly, $\rho(j) = \sigma(\tau(j))$. Because τ takes positive integer values and σ is defined on every positive integer, ρ is a function from $\mathbb{N}$ to $\mathbb{N}$. If $j < \ell$, strict increase of τ gives $\tau(j) < \tau(\ell)$. Strict increase of σ then gives

$$\rho(j) = \sigma(\tau(j)) < \sigma(\tau(\ell)) = \rho(\ell).$$

Thus ρ is strictly increasing.

3. Identify the selected values

Substituting the first selection into the second yields, for each j,

$$u_j = t_{\tau(j)} = s_{\sigma(\tau(j))} = s_{\rho(j)}.$$

The values u_j therefore select terms of s in the strictly increasing index order $\rho(1) < \rho(2) < \cdots$. Definition 11.1 says exactly that this makes u a subsequence of s.

Conclusion. The composite selection function $\sigma \circ \tau$ proves that every subsequence of a subsequence is a subsequence of the original sequence.

11.7 Exercise 11.7: A subsequence of an enumeration of all rational numbers

What we must show. If (r_n) enumerates $\mathbb{Q}$, construct a subsequence (r_{n_k}) tending to $+\infty$.

Plan. Every tail of the enumeration remains unbounded above. Prove that statement with explicit control of a finite prefix, then select successive terms that exceed their subsequence indices.

1. Show that arbitrarily large rational values occur after any prefix

Fix a positive integer N and a real threshold M. The finite prefix has a finite maximum

$$P = \max\{r_1, \ldots, r_N\}.$$

By the Archimedean property, choose a positive integer $p > \max\{M, P, 1\}$. Since $p \in \mathbb{Q}$, the enumeration contains p: there is an index j with $r_j = p$. The inequality $p > P$ prevents $j \leq N$, because every term of that prefix is at most P. Thus $j > N$ and $r_j > M$. This proves that the tail after any finite prefix is unbounded above. Before any terms are excluded, the same conclusion follows by choosing an integer $p > M$ in the enumeration.

2. Select terms recursively with increasing indices

Choose n_1 so that $r_{n_1} > 1$; this is possible by Step 1. Suppose $n_1 < \cdots < n_{k-1}$ have been chosen. Apply Step 1 to the prefix ending at n_{k-1} and to the real threshold

$$M_k = \max\{k, r_{n_{k-1}}\}.$$

It supplies an index $n_k > n_{k-1}$ with

$$r_{n_k} > M_k \geq k, \qquad r_{n_k} > r_{n_{k-1}}.$$

The resulting indices are strictly increasing, so they define a subsequence. The selected values are strictly increasing as well, although this latter property is not needed just to be a subsequence.

3. Check the definition of divergence to positive infinity

Let $M > 0$ be arbitrary. Choose an integer $K > M$. If $k > K$, the construction gives

$$r_{n_k} > k > K > M.$$

Thus the selected values eventually exceed every prescribed positive real threshold. By Definition 9.8, $r_{n_k} \to +\infty$.

Conclusion. Such a subsequence exists; the construction ensures both $n_1 < n_2 < \cdots$ and $r_{n_k} > k$, which proves its limit is $+\infty$.

11.8 Exercise 11.8: Reflection exchanges the limit inferior and limit superior

What we must show. For every real sequence (s_n), prove

$$\liminf_{n\to\infty} s_n = -\limsup_{n\to\infty}(-s_n)$$

using Definition 10.6 and the reflection identity of Exercise 5.4.

Plan. Reflect each tail before taking its extremum. If the original sequence is unbounded below, use the infinite-value conventions in Definition 10.6 directly. Otherwise all relevant tail bounds are real numbers, and their limits can be compared using finite convergence or the definition of an infinite limit.

1. Apply reflection to the nonempty tails

For each positive integer N, let

$$E_N = \{s_n : n > N\}, \qquad -E_N = \{-s_n : n > N\}.$$

Each set is nonempty, since it contains the corresponding term with index $N + 1$. Exercise 5.4 gives the extended equality

$$\inf E_N = -\sup(-E_N).$$

Write $u_N = \inf E_N$ and $v_N = \sup(-E_N)$. The equality says $u_N = -v_N$, or equivalently $v_N = -u_N$, with the negation conventions of Section 5.

2. Handle a sequence that is unbounded below

Suppose (s_n) is unbounded below. Every tail E_N is also unbounded below. Indeed, if some E_N had a real lower bound c, then

$$\min\{c, s_1, \ldots, s_N\}$$

would be a real lower bound for the entire sequence, a contradiction. Hence $u_N = -\infty$ for every N.

The reflected sequence $(-s_n)$ is unbounded above: for every real M, a term $s_n < -M$ yields $-s_n > M$. Thus $v_N = +\infty$ for every N. Definition 10.6 explicitly decrees, in these respective cases,

$$\liminf s_n = -\infty, \qquad \limsup(-s_n) = +\infty.$$

Consequently

$$-\limsup(-s_n) = -(+\infty) = -\infty = \liminf s_n.$$

This step uses the stated conventions, not a real-sequence limit law applied to a sequence of infinite symbols.

3. When the original sequence is bounded below, all these tail bounds are real

Suppose $c \in \mathbb{R}$ is a lower bound for all the s_n. Then every nonempty tail E_N is bounded below by c, so $u_N \in \mathbb{R}$ by Corollary 4.5. The reflected tail is bounded above by $-c$, so $v_N \in \mathbb{R}$ by completeness. The identity from Step 1 is now an identity of real numbers:

$$v_N = -u_N \qquad (N \in \mathbb{N}).$$

Moreover $E_{N+1} \subseteq E_N$, and restricting the set cannot decrease its infimum. Thus $u_N \leq u_{N+1}$. The real sequence (u_N) is increasing and is bounded below by c. The monotone-sequence results of Section 10 show that it either converges to a finite real number or diverges to $+\infty$. These are the only two cases to consider here.

4. Treat a finite limit of the real tail infima

If $u_N \to u \in \mathbb{R}$, then Theorem 9.2, applied to the real sequence (u_N) with scalar -1, gives $v_N = -u_N \to -u$. Definition 10.6 now identifies

$$\liminf s_n = u, \qquad \limsup(-s_n) = -u.$$

Negating the second real equality proves $-\limsup(-s_n) = u = \liminf s_n$.

5. Treat divergence of the real tail infima to positive infinity

If $u_N \to +\infty$, we show directly that $v_N = -u_N \to -\infty$. Let $M < 0$. The number $-M$ is positive, so the hypothesis supplies a threshold N_0 such that $N > N_0$ implies $u_N > -M$. Negating gives $v_N = -u_N < M$ for all such N. This is precisely the definition of $v_N \to -\infty$. Consequently Definition 10.6 gives

$$\liminf s_n = +\infty, \qquad \limsup(-s_n) = -\infty.$$

The convention $-(-\infty) = +\infty$ then proves the desired identity.

Conclusion. In all cases, $\liminf s_n = -\limsup(-s_n)$. The infinite cases follow from tail unboundedness and the infinite-limit definitions, with finite limit laws used only for real tail bounds having a finite limit.

11.9 Exercise 11.9: Closed intervals and an impossible set of subsequential limits

What we must show. (a) Show that $[a, b]$ is closed for real $a \leq b$. (b) Decide whether some real sequence can have exactly $(0, 1)$ as its set of subsequential limits.

Plan. Use the sequential meaning of closed stated with Theorem 11.9: the set contains every limit of a convergent sequence of its points. For part (b), Theorem 11.9 requires the set of subsequential limits itself to have this property.

1. Part (a): Verify the sequential closedness condition

Let (t_k) be a sequence with $t_k \in [a, b]$ for every k, and suppose $t_k \to t \in \mathbb{R}$. Thus

$$a \leq t_k \leq b \qquad (k \in \mathbb{N}).$$

If $t < a$, take $\varepsilon = (a - t)/2 > 0$. Eventually $|t_k - t| < \varepsilon$, so

$$t_k < t + \varepsilon = \frac{a + t}{2} < a,$$

contradicting $t_k \geq a$. If $t > b$, take $\varepsilon = (t-b)/2 > 0$. Eventually $|t_k - t| < \varepsilon$, so

$$t_k > t - \varepsilon = \frac{b+t}{2} > b,$$

contradicting $t_k \leq b$. Both alternatives are impossible. Hence $a \leq t \leq b$, so $t \in [a, b]$. This proves that $[a, b]$ is closed in the required sequential sense.

The argument includes $a = b$: in that case all terms equal a and the only possible limit is a. If extended limits are also considered, neither infinite symbol can arise from such a bounded sequence. For $+\infty$, the positive threshold $\max\{1, b+1\} > b$ cannot be exceeded; for $-\infty$, the negative threshold $\min\{-1, a-1\} < a$ cannot be crossed downwards.

2. Part (b): Apply Theorem 11.9 to a sequence of proposed limits

Suppose, for a contradiction, that a real sequence (s_n) has exactly $S = (0, 1)$ as its set of subsequential limits. Define

$$t_k = \frac{1}{k+1} \qquad (k \in \mathbb{N}).$$

Since $k + 1 \geq 2$, we have $0 < t_k \leq 1/2 < 1$. Thus $t_k \in S$ for every k. Also $t_k \to 0$: for any $\varepsilon > 0$, choose an integer $K > 1/\varepsilon$; then $k > K$ gives

$$|t_k - 0| = \frac{1}{k+1} < \frac{1}{k} < \varepsilon.$$

Theorem 11.9 applies to this sequence of finite members of S and forces its limit 0 to belong to S. But $0 \notin (0, 1)$, a contradiction.

Conclusion. (a) Every closed interval $[a, b]$, including a one-point interval, is closed in the sequential sense. (b) No sequence has $(0, 1)$ as exactly its set of subsequential limits.

11.10 Exercise 11.10: The alternating diagonal traversal in Figure 11.2

What we must show. For the sequence listed by the arrows of Figure 11.2, find (a) its exact set of subsequential limits and (b) its limit superior and limit inferior.

Plan. The figure has the same row repeated infinitely many times. Each reciprocal value therefore occurs at arbitrarily large indices. The top row also supplies a subsequence of values tending to 0. To prove completeness of the answer, isolate any other positive proposed limit from all reciprocal values.

1. Describe the array and its order precisely

Number rows and columns by $i, j \in \mathbb{N}$. The entry in position (i, j) is $1/j$, so the array is

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

The traversal groups the entries by diagonals $i + j = m + 1$, with $m = 1, 2, 3, \ldots$. Diagonal m has exactly m entries. For odd m, the traversal lists columns $1, 2, \ldots, m$; for even m, it lists columns

$m, m-1, \ldots, 1$. Thus the successive diagonal blocks begin

$$\underbrace{1}_{m=1},\quad \underbrace{\tfrac{1}{2}, 1}_{m=2},\quad \underbrace{1, \tfrac{1}{2}, \tfrac{1}{3}}_{m=3},\quad \underbrace{\tfrac{1}{4}, \tfrac{1}{3}, \tfrac{1}{2}, 1}_{m=4},\quad \ldots.$$

This order agrees with the alternating directions of the arrows.

2. Check explicitly that choosing later diagonals gives later indices

Before diagonal m there are

$$T_{m-1} = 1 + 2 + \cdots + (m-1) = \frac{m(m-1)}{2}$$

listed entries, with $T_0 = 0$. For $m \geq j$, the occurrence of $1/j$ in diagonal m has index

$$I(m, j) = \begin{cases} \dfrac{m(m-1)}{2} + j, & m \text{ odd}, \\ \dfrac{m(m-1)}{2} + (m - j + 1), & m \text{ even}. \end{cases}$$

Indeed, its position within that diagonal is j when columns are in increasing order, and $m - j + 1$ when they are in decreasing order. In both cases

$$T_{m-1} + 1 \leq I(m, j) \leq T_m, \qquad T_m = \frac{m(m+1)}{2}.$$

Every index of diagonal $m+1$ is at least $T_m + 1$, which is strictly larger than every index of diagonal m. Consequently selections made from successive diagonals always have strictly increasing indices.

3. Part (a): Obtain every reciprocal as a subsequential limit

Fix $j \in \mathbb{N}$. Every diagonal $m \geq j$ contains column j once, at the matrix position $(m + 1 - j, j)$. Choose the indices

$$n_k = I(j + k - 1, j) \qquad (k \in \mathbb{N}).$$

They are strictly increasing by Step 2, and the corresponding sequence values satisfy $s_{n_k} = 1/j$ for every k. This constant subsequence converges to $1/j$. Thus every member of $\{1/j : j \in \mathbb{N}\}$ is a subsequential limit.

4. Part (a): Obtain zero from the top row

The top-row entry in column k lies in diagonal k and has value $1/k$. Select it at index

$$n_k = I(k, k) = \begin{cases} \dfrac{k(k+1)}{2}, & k \text{ odd}, \\ \dfrac{k(k-1)}{2} + 1, & k \text{ even}. \end{cases}$$

These indices are strictly increasing because they come from successive diagonals. For example, they begin $1, 2, 6, 7, 15, \ldots$. The selected values are exactly $s_{n_k} = 1/k$. Given $\varepsilon > 0$, an integer $K > 1/\varepsilon$ ensures $k > K$ implies $|1/k - 0| < \varepsilon$. Therefore $s_{n_k} \to 0$, and 0 is also a subsequential limit.

5. Part (a): Exclude every other proposed limit

Every term of the full sequence belongs to $D = \{1/j : j \in \mathbb{N}\} \subseteq (0, 1]$. Hence any finite subsequential limit is in $[0, 1]$, by the same eventual-bound argument used in Exercise 11.5. The bounds also rule out both infinite subsequential limits.

Suppose $t > 0$ and $t \notin D$. We show that a fixed positive neighborhood of t contains no term of the sequence. Choose an integer $J > \max\{2, 2/t\}$. For every $j \geq J$, we have

$$\frac{1}{j} \leq \frac{1}{J} < \frac{t}{2}, \qquad \left|\frac{1}{j} - t\right| = t - \frac{1}{j} > \frac{t}{2}.$$

For the finitely many indices $1 \leq j < J$, none of the values $1/j$ equals t. Their distances from t are strictly positive, so the finite minimum

$$d = \min_{1 \leq j < J} \left|\frac{1}{j} - t\right|$$

exists and is positive. Define $\delta = \min\{t/4, d/2\} > 0$. If $j \geq J$, then $|1/j - t| > t/2 > \delta$. If $j < J$, then $|1/j - t| \geq d > \delta$. Thus every term of the original sequence satisfies $|s_n - t| > \delta$. No subsequence can converge to t, since convergence would require its terms eventually to have distance less than δ from t.

Every possible finite limit is either 0, a member of D, or a positive number not in D. Steps 3–5 account for all these possibilities, and the infinite cases were excluded. Therefore

$$S = \{0\} \cup \left\{\frac{1}{j} : j \in \mathbb{N}\right\}$$

is exactly the set of subsequential limits.

6. Part (b): Use the extreme members of the computed set

The set S has minimum 0 and maximum 1. Theorem 11.8(ii) therefore gives

$$\limsup_{n\to\infty} s_n = \sup S = 1, \qquad \liminf_{n\to\infty} s_n = \inf S = 0.$$

Conclusion. (a) The exact set is $S = \{0\} \cup \{1/j : j \in \mathbb{N}\}$. (b) The limit superior is 1 and the limit inferior is 0.

11.11 Exercise 11.11: An increasing sequence approaching a supremum from the set

What we must show. For a nonempty bounded set $S \subseteq \mathbb{R}$, construct an increasing sequence (s_n) with every $s_n \in S$ and $s_n \to \sup S$.

Plan. If the supremum belongs to the set, use the constant sequence at that point. Otherwise, at each stage choose a point of S above both the previous value and a closer lower approximation to the supremum.

1. State the necessary nonemptiness condition and the meaning of increasing

The exercise's statement needs the implicit assumption $S \neq \varnothing$. For the empty set there is no sequence of points in S, and its supremum is not defined by the real-supremum conventions being used. We therefore assume that S is nonempty and bounded. In fact, the construction needs only that it be bounded above. Completeness gives the finite real number

$$M = \sup S.$$

As in Section 10 and in the note to this exercise, increasing means nondecreasing: $s_n \leq s_{n+1}$. A constant sequence is allowed. When $M \notin S$, the construction below will give strict increase.

2. The case in which the supremum belongs to the set

If $M \in S$, define $s_n = M$ for every $n \in \mathbb{N}$. Then all terms belong to S and $s_n = s_{n+1}$, so the sequence is increasing in the stated sense. For every $\varepsilon > 0$,

$$|s_n - M| = 0 < \varepsilon$$

at every index. Hence $s_n \to M$.

3. Begin the construction when M is not a member

Suppose $M \notin S$. Since M is an upper bound, every $s \in S$ satisfies $s \leq M$; the nonmembership of M strengthens this to $s < M$ for every $s \in S$. The number $M - 1 < M$ cannot be an upper bound for S, because M is its least upper bound. Therefore some $s_1 \in S$ satisfies

$$M - 1 < s_1 < M.$$

4. Continue the construction with an explicitly admissible threshold

Suppose $s_{n-1} \in S$ has been chosen, where $n \geq 2$. Both $s_{n-1} < M$ and $M - 1/n < M$ hold. Therefore

$$c_n = \max\left\{s_{n-1}, M - \frac{1}{n}\right\} < M.$$

As $c_n < M = \sup S$, the number c_n is not an upper bound for S. Choose $s_n \in S$ with $s_n > c_n$. Every point of S is less than M, so this choice satisfies

$$s_{n-1} < s_n < M, \qquad M - \frac{1}{n} < s_n < M.$$

The procedure can be repeated at every positive integer stage. It gives a sequence in S that is strictly increasing. The displayed approximation bound also holds at $n = 1$ by Step 3.

5. Verify convergence to the supremum

For every n, the construction gives

$$0 < M - s_n < \frac{1}{n}.$$

Let $\varepsilon > 0$. Choose an integer $N > 1/\varepsilon$ by the Archimedean property. If $n > N$, then

$$|s_n - M| = M - s_n < \frac{1}{n} < \varepsilon.$$

Thus $s_n \to M$ by the definition of convergence.

Conclusion. With the necessary hypothesis $S \neq \varnothing$, an increasing sequence in S converging to $\sup S$ always exists. It may be taken constant when $\sup S \in S$, and strictly increasing when $\sup S \notin S$.